

Troubleshooting Guide: Crude Preheat Train — Progressive Exchanger Fouling Reducing Furnace Inlet Temperature

Category: Heat Transfer Equipment / Predictive Maintenance **Unit:** Crude Preheat Train (series of shell-and-tube exchangers upstream of the atmospheric heater)
Tools: HTRI/EDR-based exchanger performance monitoring, network-level U-value trending **Fluid Package:** Not applicable in the VLE sense — HTRI/EDR use built-in crude/hydrocarbon property correlations for thermal-hydraulic rating, similar to Case Study 6 **Symptom:** Furnace inlet (crude preheat) temperature gradually declining, forcing higher furnace duty and fuel consumption

Note on case type: Like the seawater cooler case (Case Study 6), this is a **predictive/quantitative trending** exercise focused on identifying *which* exchanger in a train is degrading and *why*, rather than distinguishing between unrelated root-cause hypotheses.

1. Quick Reference

Item	Detail
Reported symptom	Furnace inlet (crude preheat) temperature gradually dropping over time; furnace duty and fuel use rising to compensate
Task	Identify which exchanger(s) in the train were fouling and why, to target corrective action instead of blanket cleaning the whole train
Actual root cause	Asphaltene precipitation in a specific crude/VGO exchanger, caused by incompatibility between blended crude sources
Fix	Mechanical cleaning of the affected exchanger during a turnaround; review of crude blending compatibility
Diagnostic signal	HTRI-based U-value trending isolated the fouling to one specific exchanger in the train, rather than the whole network degrading uniformly
Prevention	Periodic U-value trending per exchanger; asphaltene/blending compatibility screening before crude source changes

2. Symptom

- **Furnace inlet temperature (crude preheat outlet) gradually declined over time.**
- **Furnace duty and fuel consumption increased** to compensate and hold the same furnace outlet/process target temperature.

3. Why This Needed Per-Exchanger Analysis, Not a Blanket Response

A preheat train is a series of multiple exchangers. A declining overall preheat temperature could result from **one badly fouled exchanger**, or from **several exchangers fouling moderately** — these call for very different responses (targeted cleaning of one unit vs. a broader train-wide cleaning campaign). Cleaning the wrong exchanger, or the whole train unnecessarily, wastes turnaround time and cost. The goal was to isolate **which** exchanger(s) were responsible before committing cleaning resources.

4. Diagnostic Approach

Step 1 — Model the preheat train and calculate actual U-values per exchanger

Each exchanger in the train was modeled/monitored in HTRI (or equivalent EDR tool), using actual operating data to calculate the **actual overall heat transfer coefficient (U)** for each unit individually, not just the train's aggregate performance.

Same core technique as Case Study 6: compare actual U to clean/design U, per exchanger, and trend it over time — this is a thermal-hydraulic rating exercise using built-in property correlations, not a VLE flash calculation.

Step 2 — Trend U-value per exchanger over time

Historical trending of U for each individual exchanger identified that the **decline was concentrated in one specific crude/VGO exchanger**, while the other exchangers in the train remained close to their clean/design U values.

Step 3 — Investigate the specific fouling mechanism at that exchanger

With the fouling isolated to one unit, the investigation turned to *why that specific exchanger*. Reviewing recent operating history alongside the crude supply record identified a period of **crude blending from multiple sources**.

Step 4 — Confirm asphaltene incompatibility as the mechanism

Crude compatibility (asphaltene stability) analysis confirmed that the **blended crude sources were incompatible**, causing asphaltenes to destabilize and precipitate — depositing preferentially in the exchanger(s) operating in the temperature/velocity range where that particular crude blend's asphaltenes were least stable.

Quantitative Basis

- Train of 6 exchangers; clean/design U values range 85–130 Btu/hr·ft²·°F depending on service.
- Furnace inlet temperature: normal baseline 546°F, dropped to **498°F over 5 months — a 48°F loss**.
- U-trending isolated the decline: **5 of 6 exchangers held within 8% of clean design U**; the crude/VGO exchanger (E-104) dropped from a clean 105 to **61 Btu/hr·ft²·°F — a 42% loss**.
- Crude compatibility screening (ASTM D7157-style P-value) on the blended crude measured a **P-value of 1.02**, below the 1.1 minimum threshold for asphaltene stability — confirming incompatibility following introduction of a new crude source.

5. Root Cause

Blending of incompatible crude sources caused asphaltene destabilization and precipitation, which deposited and fouled a specific crude/VGO exchanger in the preheat train — reducing that exchanger’s heat transfer performance and, cumulatively, the overall furnace inlet temperature, even though the rest of the train remained close to design performance.

6. Corrective Action

1. **Mechanically cleaned the affected exchanger** during a scheduled turnaround.
2. **Reviewed crude blending compatibility practices**, incorporating asphaltene stability screening before future crude source changes/blends.

7. Verification

- E-104 U-value returned to **98 Btu/hr·ft²·°F — 93% of clean design**, following mechanical cleaning.
- Furnace inlet temperature recovered to **542°F**, and furnace fuel gas consumption dropped **~9%** back to its pre-fouling baseline.
- Held stable over the following **30 days**.

8. Prevention / Long-Term Fix

- Established **periodic U-value trending per exchanger** across the preheat train, so future fouling can be isolated to a specific unit early, rather than discovered only as an aggregate furnace-duty symptom.
- Added **asphaltene/blending compatibility screening** as a standard check before introducing a new crude blend, addressing the trigger mechanism rather than only the resulting fouling.

9. General Troubleshooting Checklist (Reusable)

- ☐ Confirm the symptom (declining preheat temperature, rising furnace duty/fuel use) with trend data over time, not a single reading
- ☐ Model/monitor **each exchanger in the train individually**, not just the train’s aggregate outlet temperature
- ☐ Calculate and trend **actual U vs. clean/design U per exchanger** to isolate which unit(s) are fouling
- ☐ Once isolated, review recent operating and feed history specific to that exchanger’s service (crude source, blend changes, temperature/velocity regime)
- ☐ If crude blending is involved, check **asphaltene stability/compatibility** between the blended sources
- ☐ Target cleaning to the specific exchanger(s) identified, rather than cleaning the whole train by default
- ☐ After cleaning, confirm U recovery at the specific exchanger AND overall furnace inlet temperature/duty recovery
- ☐ Add compatibility screening to the crude blending change-management process to prevent recurrence

10. Key Takeaway

A declining preheat train is a network problem, but the fix is almost always at a **specific exchanger**, not the whole train. Trend U-value per exchanger, not just the aggregate outlet temperature, to isolate where fouling is actually occurring — then investigate what changed *for that unit specifically* (often a feed/blend change) rather than assuming generic fouling. When crude blending is involved, treat asphaltene compatibility as something to check proactively before blending, not something to diagnose after a preheat train has already fouled.

Related Concepts / Tags

crude-preheat-train exchanger-fouling asphaltene-precipitation crude-compatibility
HTRI U-value-trending furnace-inlet-temperature predictive-maintenance shell-and-tube

This guide is derived from a representative field troubleshooting scenario. Values and thresholds shown are illustrative and should be validated against your own unit's design basis before applying.